Weitong Long

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Research Interests

- Sustainable food system transformations
- Global food supply chains, international trade, and environmental spillovers
- Climate change mitigation and air and water pollution control
- Interactions among food security, environmental sustainability, and human health
- Integrated environmental-economic-health modelling of food systems

Professional Experience

- **College of Environmental Sciences and Engineering, Peking University** *04/2026-03/2028 (expected)*
  Postdoctoral Researcher with the Boya Postdoctoral Fellowship Beijing, China
  ◦ **Supervisor:** Dr. Hancheng Dai

Education

- **Wageningen School of Social Sciences, Wageningen University & Research** *09/2020-01/2026*
  PhD of Economics in Environmental Economics and Natural Resources Wageningen, the Netherlands
  ◦ **Supervisor:** Dr. Xueqin Zhu, Dr. Hans-Peter Weikard, Prof. Dr. Oene Oenema, and Prof. Dr. Yong Hou
  ◦ **Program:** The Sino-Dutch Agriculture Green Development (AGD) PhD program [Link]
  ◦ **Dissertation:** Towards sustainable food systems in China: insights from an integrated environmental-economic modelling approach *(submitted in September 2025 and defended in January 2026)* [Link]
- **School of Veterinary Medicine, University of California, Davis** *10/2024-01/2025*
  Visiting PhD Student Davis, the United States
  ◦ **Supervisor:** Dr. Luis M. Peña-Lévano
- **College of Resources and Environmental Sciences, China Agricultural University** *09/2020-08/2021*
  Visiting PhD Student Beijing, China
  ◦ **Supervisor:** Prof. Dr. Yong Hou
- **College of Resources and Environmental Sciences, China Agricultural University** *09/2018-06/2020*
  Master of Agriculture in Plant Nutrition Beijing, China
  ◦ **Supervisor:** Prof. Dr. Yong Hou and Dr. Hongliang Wang
  ◦ **Dissertation:** Nitrogen footprint of China's pig production and mitigation measures through feed management
- **College of Resources, Hunan Agricultural University** *09/2014-06/2018*
  Bachelor of Agriculture in Agricultural Resources and Environment Changsha, China
  ◦ **Dual Bachelor:** Dual Bachelor of Arts in English

Training Courses

- **International Summer School on Geograohy and Sustainability** *08/2026*
  Michigan State University, University of Hong Kong & Beijing Normal University Beijing, China
  ◦ **Course:** Frontiers in Telecoupling and Metacoupling Research for Addressing Global Sustainability Challenges
- **Land Management for Sustainability (LAMASUS) Summer School** *06/2026*
  International Institute of Applied Systems Analysis (IIASA) Laxenburg, Austria
  ◦ **Course:** Advanced Modelling Techniques for Land-use and Policy Analysis
- **European Association of Environmental and Resource Economists (EAERE) Summer School** *07/2023*
  University of Graz Graz, Austria
  ◦ **Course:** Transnational and Cascading Climate Risks and Adaptation
- **Dynamic General Equilibrium Modelling Course** *07/2021*
  Victoria University & University of International Business and Economics Beijing, China
  ◦ **Course:** CHINAGEM, A Monash-Styled Dynamic Computable General Equilibrium Model of China

Publications

F=First Author, O=Other

equal contribution; * corresponding author
Citations (Google Scholar: October 28, 2026): Total = 415; H-index = 9; I10-index = 9

[F-4] **Long, W.**, Zhu, X.* (2025). Upcycling of food waste and food processing by-products into animal feed is not a panacea. In: ***Nature Food (SCI Q1; IF=24.7; Featured Research Briefing)***, 6(9), 835–836. DOI: 10.1038/s43016-025-01235-7 [Link]

[F-3] **Long, W.**, Zhu, X.*, Weikard, H.P., Oenema, O., Hou, Y.* (2025). Rebound effects may undermine the benefits of upcycling food waste and food processing by-products as animal feed in China. In: ***Nature Food (SCI Q1; IF=24.7; Research Article)***, 6(9), 881–891. DOI: 10.1038/s43016-025-01219-7 [Link]

[F-2] **Long, W.**, Zhu, X.*, Weikard, H.P., Oenema, O., Hou, Y.* (2024). Exploring sustainable food system transformation options in China: An integrated environmental-economic modelling approach based on the applied general equilibrium framework. In: ***Sustainable Production and Consumption (SCI & SSCI Q1; IF=9.2)***, 51, 42-54. DOI: 10.1016/j.spc.2024.09.004 [Link]

[F-1] **Long, W.#**, Wang, H.#, Hou, Y.*, Chadwick, D., Ma, Y., Cui, Z., & Zhang, F. (2021). Mitigation of multiple environmental footprints for China's pig production using different land use strategies. In: ***Environmental Science & Technology (SCI Q1; IF=12.2)***, 51, 42-54. DOI: 10.1021/acs.est.0c08359 [Link]

[O-8] Tan, M., Hou, Y.*, Zhang, T., Ma, Y., **Long, W.**, Gao, C., Liu, P., Fang, Q., Dai, G., Shi, S., Reijneveld, A., & Oenema, O. (2023). Relationships between livestock density and soil phosphorus contents–County and farm level analyses. In: ***Catena (SCI Q1; IF=6.6)***, 222, 106817. DOI: 10.1016/j.catena.2022.106817 [Link]

[O-7] Tan, M., Hou, Y.*, Zhang, L., Shi, S., **Long, W.**, Ma, Y., Zhang, T., & Oenema, O. (2023). Decision-making environment of low-protein animal feeding in dairy and poultry farms in China. In: ***Nutrient Cycling in Agroecosystems (SCI Q2; IF=3.2)***, 127(1), 85-96. DOI: 10.1007/s10705-023-10295-9 [Link]

[O-6] Tan, M., Hou, Y.*, Zhang, L., Shi, S., **Long, W.**, Ma, Y., Zhang, T., & Oenema, O. **Long, W.**, Velthof, G. L., ... & Zhang, F. (2022). Nutrient use efficiency of intensive dairy farms in China–Current situation and analyses of options for improvement. In: ***Agricultural Systems (SCI Q1; IF=6.2)***, 203, 103495. DOI: 10.1016/j.agsy.2022.103495 [Link]

[O-5] Tong, B., Zhang, L., Hou, Y.*, Oenema, O., **Long, W.**, Velthof, G. L., ... & Zhang, F. (2022). Lower pork consumption and technological change in feed production can reduce the pork supply chain environmental footprint in China. In: ***Nature Food (SCI Q1; IF=24.7)***, 1-10. DOI: 10.1038/s43016-022-00640-6 [Link]

[O-4] Ma, Y., Hou, Y.*, Dong, P., Velthof, G. L., **Long, W.**, Ma, L., Ma, W., Jiang, R., & Oenema, O. (2022). Cooperation between specialized livestock and crop farms can reduce environmental footprints and increase net profits in livestock production. In: ***Journal of Environmental Management (SCI Q2; IF=9.2)***, 302, 113960. DOI: 10.1016/j.jenvman.2021.113960 [Link]

[O-3] Wang, H., **Long, W.**, Chadwick, D., Zhang, X., Zhang, S., Piao, X.*, & Hou, Y.* (2022). Dietary acidifiers as an alternative to antibiotics for promoting pig growth performance: A systematic review and meta-analysis. In: ***Animal Feed Science and Technology (SCI Q2; IF=3.2)***, 115320. DOI: 10.1016/j.anifeedsci.2022.115320 [Link]

[O-2] Tan, M., Hou, Y.*, Zhang, L., Shi, S., **Long, W.**, Ma, Y., Zhang, T., Li, F., & Oenema, O. (2021). Operational costs and neglect of end-users are the main barriers to improving manure treatment in intensive livestock farms. In: ***Journal of Cleaner Production (SCI Q1; IF=10.7)***, 289, 125149. DOI: 10.1016/j.jclepro.2020.125149 [Link]

[O-1] Wang, H., **Long, W.**, Chadwick, D., Velthof, G. L., Oenema, O., Ma, W., Wang, J., Qin, W., Hou, Y.*, & Zhang, F. (2020). Can dietary manipulations improve the productivity of pigs with lower environmental and economic cost? A global meta-analysis. In: ***Agriculture, Ecosystems & Environment (SCI Q1; IF=7.7)***, 289, 106748. DOI: 10.1016/j.agee.2019.106748 [Link]

Working Papers

F=First Author, O=Other

equal contribution; * corresponding author

[F-2] **Long, W.**, Zhu, X.*, Hou, Y.*, Peña-Lévano, L. M., Garcia-Covarrubias L., Boy, K.-F. (2026). Trade-induced land-use change may reshape environmental performance of China's dietary and afforestation policies with implications for a 2 °C target. [*Full paper available upon request*] (*Submitted*).

[F-1] Lan, Y.#, **Long, W.#**, Zamani, O., Chai, L., Zhuang, M., Fu, B., Yang, Y., Liu, G.G., Xie, Y.*, Dai, H.* (2026). Joint fertilizer and food trade disruptions at the Strait of Hormuz amplify global food insecurity. [*Full paper available upon request*] (*Submitted*).

Conference Presentations

**** indicates presenter***

[14] **Long, W.***, Yin, C., Ni, Z., Lan, X., Li, Z., Ni, X., Yao, S. (08/2026). Metacoupled impacts of a global dietary shift on environmental sustainability and human health. An oral presentation delivered at the *International Summer School on Geograohy and Sustainability*, Beijing, China.

[13] **Long, W.**, Zhu, X., Hou, Y., Peña-Lévano, L. M.*, Garcia-Covarrubias L., Boy, K.-F. (07/2026). Emission leakages through trade-induced land use changes may undermine the emission mitigation effectiveness of dietary shifting and afforestation policy in China. An oral presentation delivered at the *2026 Agricultural & Applied Economics Association (AAEA) Annual Meeting*, Kansas City, Missouri, the United States.

[12] **Long, W.**, Zhu, X.*, Hou, Y., Peña-Lévano, L. M., Garcia-Covarrubias L., Boy, K.-F. (07/2026). Emission leakages through trade-induced land use changes may undermine the emission mitigation effectiveness of dietary shifting and afforestation policy in China. An oral presentation delivered at the *7th World Congress of Environmental and Resource Economists (WCERE)*, Carcavelos, Portugal.

[11] **Long, W.**, Zhu, X., Hou, Y., Peña-Lévano, L. M.*, Garcia-Covarrubias L., Boy, K.-F. (06/2026). Emission leakages through trade-induced land use changes may undermine the emission mitigation effectiveness of dietary shifting and afforestation policy in China. An oral presentation to be delivered at the *International Food and Agribusiness Management Association (IFAMA) 2026 World Conference*, Cork, Ireland.

[10] **Long, W.***, Zhu, X., Hou, Y., Peña-Lévano, L. M., Garcia-Covarrubias L., Boy, K.-F. (06/2026). Emission leakages through trade-induced land use changes may undermine the emission mitigation effectiveness of dietary shifting and afforestation policy in China. A poster presentation to be delivered at the *Land Management for Sustainability (LAMASUS) Summer School 2026*, International Institute of Applied Systems Analysis (IIASA), Laxenburg, Austria.

[9] **Long, W.**, Zhu, X., Hou, Y., Peña-Lévano, L. M.*, Garcia-Covarrubias L., Boy, K.-F. (08/2025). Land-use emission leakages from China's dietary shift and afforestation amplify food insecurity and economic losses under the 2 °C target. A poster presentation delivered at the *XVIII European Association of Agricultural Economists (EAAE) Congress*, Bonn, Germany.

[8] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (02/2025). Quantifying the environmental and economic impacts of upcycling food waste and food processing by-products as animal feed: a general equilibrium approach. An oral presentation delivered at the *4th Dutch Environmental and Resource Economics (DEARE) Day workshop*, Wageningen, the Netherlands.

[7] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (07/2024). Quantifying the environmental and economic impacts of feeding China's monogastric livestock with food waste: a general equilibrium approach. An oral presentation delivered at the *29th Annual Conference of the European Association of Environmental and Resource Economists (EAERE)*, Leuven, Belgium.

[6] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (06/2024). Quantifying the environmental and economic impacts of feeding China's monogastric livestock with food waste: a general equilibrium approach. An oral presentation delivered at the *III Economy for The Common Good International Conference (ECGIC)*, Leeuwarden, Fryslân, the Netherlands.

[5] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (05/2024). The asymmetric impacts of feeding China's monogastric livestock with food waste on food security and environment sustainability. An oral presentation delivered at the *9th Sino-Dutch Agriculture Green Development (AGD) Symposium*, Wageningen University & Research, Wageningen, the Netherlands.

[4] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (08/2023). Integrated Environmental-economic modelling of sustainable food systems in China. A poster presentation delivered at the *XVII European Association of Agricultural Economists (EAAE) Congress*, Rennes, France.

[3] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (07/2023). Integrated Environmental-economic modelling of sustainable food systems in China. An oral presentation delivered at the *European Association of Environmental and Resource Economists (EAERE) Summer School 2023*, University of Graz, Graz, Austria.

[2] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (02/2023). Environmental trade-offs of dietary structure change can be alleviated by cleaner technology and emission restriction. An oral presentation delivered at the *7th Sino-Dutch Agriculture Green Development (AGD) Symposium*, Wageningen University & Research, Wageningen, the Netherlands.

[1] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (10/2022). An environmental-economic framework for assessing the impacts of adjustments in crop and livestock systems. An oral presentation delivered at the *Wageningen School of Social Sciences (WASS) PhD Day 2022*, Wageningen University & Research, Wageningen, the Netherlands.

SEMINAR TALKS

**** indicates presenter***

[5] **(Invited) Long, W.*** (06/2026). Challenges and opportunities of food system transformations: Growth and insights from the Sino-Dutch Agriculture Green Development (AGD) PhD program. Oral presentation delivered at the *Plant Nutrition Weekly Seminar*, China Agricultural University, Beijing, China (Online).

[4] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (04/2024). Quantifying the environmental and economic impacts of feeding China's monogastric livestock with food waste: a general equilibrium approach. Oral presentation delivered at the *EconMonday Weekly Lunch Seminar*, Wageningen University & Research, Wageningen, the Netherlands.

[3] **(Invited) Long, W.*** (12/2023). Food system environmental policy analysis and method application. Oral presentation delivered at the *Plant Nutrition Weekly Seminar*, China Agricultural University, Beijing, China (Online).

[2] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (06/2023). Exploring options for sustainable food systems in China: An integrated environmental-economic modelling approach. Oral presentation delivered at the *EconMonday Weekly Lunch Seminar*, Wageningen University & Research, Wageningen, the Netherlands.

[1] **Long, W.***, Zhu, X., Weikard, H.P., Oenema, O., Hou, Y. (11/2022). The global environmental consequences of adjustments in the food systems in China. Oral presentation delivered at the *EconMonday Weekly Lunch Seminar*, Wageningen University & Research, Wageningen, the Netherlands.

Grants and Awards

- The First Place Best Paper Award at the IFAMA 2026 World Conference (1st out of 440+ papers) *06/2026*
- Boya Postdoctoral Fellowship at Peking University (380 fellowships each year) *12/2025*
- Junior Researcher Grant from WASS for the four-month research at UC Davis (4,000 €; PI) *07/2024*
- Travel Grant from the LEB Travel Fund to participate in the XVII EAAE Congress (750 €; PI) *06/2023*
- Honorarium for co-organising the 6th and 7th Sino-Dutch AGD Symposiums (1,000 €; PI) *02/2023*
- Educational backpack from WASS for taking courses and attending conferences (3,500 €; PI) *02/2021*
- Research Grant from the Sino-Dutch AGD PhD Program for data collection (8,300 €; PI) *12/2020*
- PhD Full Scholarship from China Scholarship Council (CSC) for PhD study at Wageningen University *12/2020*
- Excellent Master's Degree Thesis from the Chinese Society of Plant Nutrition and Fertiliser Science (Awarded to the Top 1% best master thesis in China) *08/2020*
- The First Class Master Academic Scholarship at China Agricultural University (Top 1%) *10/2019*
- The First Prize of English Speech Contest at China Agricultural University (Top 1%) *11/2018*

Teaching Experience

- **ENR32806: Economic Modelling of Sustainability Challenges (Master level, 6 ECTS)** *2023 & 2024 Spring*
  Wageningen University & Research Wageningen, the Netherlands
  - Assisted in teaching ENR32806: Economic Modelling of Sustainability Challenges (Master level, 6 ECTS) with Dr. Xueqin Zhu and Dr. Jack Peerlings
  - Organised tutorials, provided support to master students with modelling and coding inquiries, and completed grading assignments
- **ENR22806: Principles of Climate Change Economics and Policy (Master level, 6 ECTS)** *2022 Winter*
  Wageningen University & Research Wageningen, the Netherlands
  - Assisted in teaching ENR22806: Principles of Climate Change Economics and Policy (Master level, 6 ECTS) with Dr. Xueqin Zhu and Dr. Ina Möller
  - Provided feedback on literature review papers of master students and completed grading assignments

Mentorinng Experience

- **Co-supervisor of Master Thesis** *03/2022-05/2024*
  Wageningen University & Research Wageningen, the Netherlands
  - Co-supervised Jia Zhou with Dr. Xueqin Zhu on the master thesis of "Exploring optimal cover crop management practice in China Loess Plateau by model simulation and mathematical programming"
  - Co-supervised Huangshu Zhao with Dr. Hans-Peter Weikard on the master thesis of "Optimising county-level manure redistribution in Handan, China to balance economic and environmental benefits"
  - Co-supervised Kehan Qiu with Dr. Rolf Groeneveld on the master thesis of "A computable general equilibrium model for evaluating the economic impact of biofuel policy in the Netherlands"
- **Chair of Master Thesis Ring** *01/2022-12/2022*
  Wageningen University & Research Wageningen, the Netherlands
  - Organised weekly sessions to help master students improve the clarity and conciseness of their thesis
  - Facilitated constructive peer feedback to enhance the quality of master students' written work

Academic Services

- **Early Career Editorial Board Member:** Carbon Footprints (2026–2028)
- **Peer-reviewed Journal Article Reviewer:** Nature Food (2025)
- **Conference Parallel Session Chair and Discussant:** The 29th Annual Conference of the EAERE (2024)
- **Conference Co-organiser:** The 6th and 7th Sino-Dutch AGD Symposiums (2022-2023)
- **Conference Paper Reviewer:** The XVIII EAAE Congress (2025)
- **Conference Abstract Reviewer:** The 2024, 2025, and 2026 AAEA Annual Meetings (2024-2026)

Skills

- **Programming:** General Algebraic Modeling System (GAMS, advanced, e.g. model establishment), General Equilibrium Modelling PACKage (GEMPACK, intermediate), and R (intermediate)

- **Modelling:** Applied general equilibrium (AGE) modelling, life cycle assessment (LCA), input-output (I-O) analysis, material flow analysis (MFA), and meta-analysis
- **Software:** ArcGIS, Simapro, Github, Latex, and Microsoft Office
- **Languages:** Chinese (Native), English (Strong reading, writing, and speaking competencies), IELTS score (7.0)

Professional Association Memberships

- American Economic Association (AEA)
- Agricultural & Applied Economics Association (AAEA)
- Association of Environmental and Resource Economists (AERE)
- European Economic Association (EEA)
- European Association of Agricultural Economists (EAAE)
- European Association of Environmental and Resource Economists (EAERE)
- American Geosciences Union (AGU)
- European Geosciences Union (EGU)
- International Society for Industrial Ecology (ISIE)
- Global Trade Analysis Project (GTAP) Network

References

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